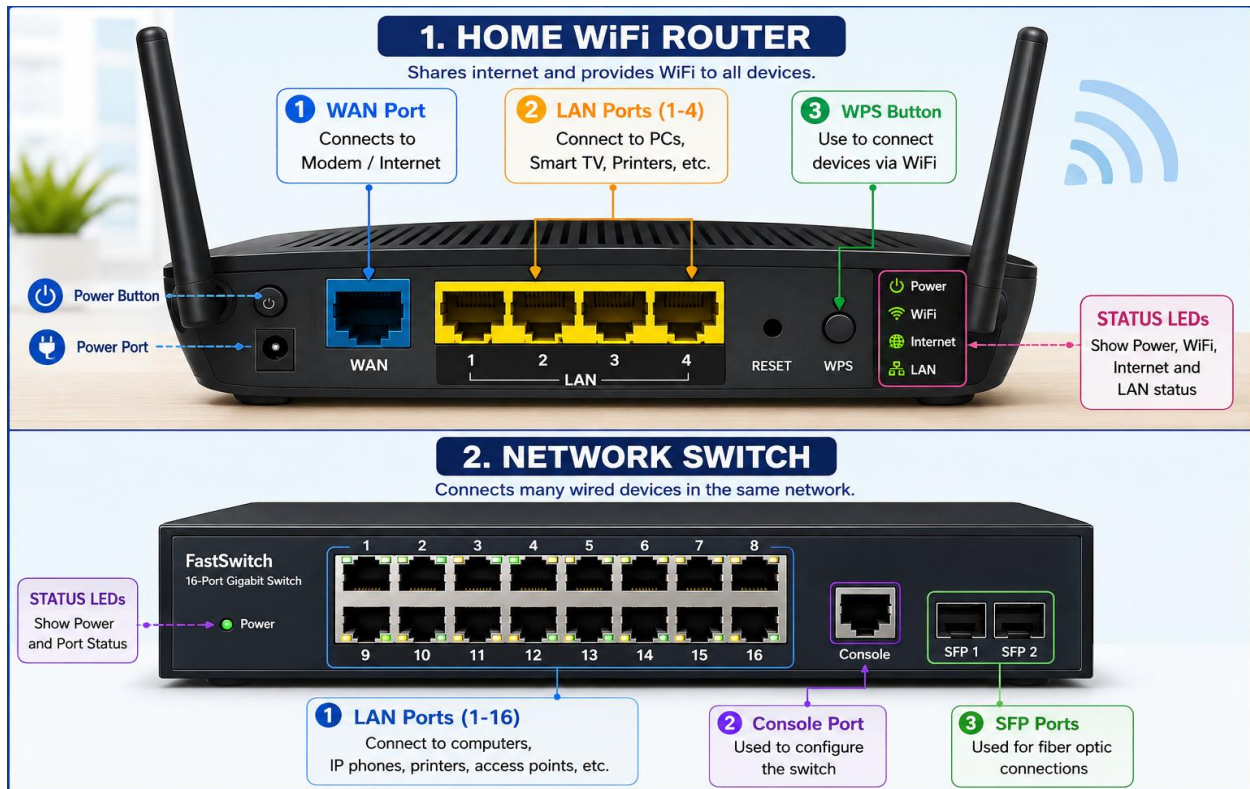


1. Draw a labelled diagram of a typical home WiFi router and a network switch, clearly marking at least 3 different ports (such as WAN, LAN, Console) and the location of status LEDs



EXPLAINED ::

- This is a typical home WiFi router and a network switch.
- The WAN port connects the router to the modem and the internet.
- The LAN ports connect computers, printers, and other wired devices.
- The Console port is used to configure the network switch.
- The Status LEDs show the power, internet, WiFi, and LAN connection status.

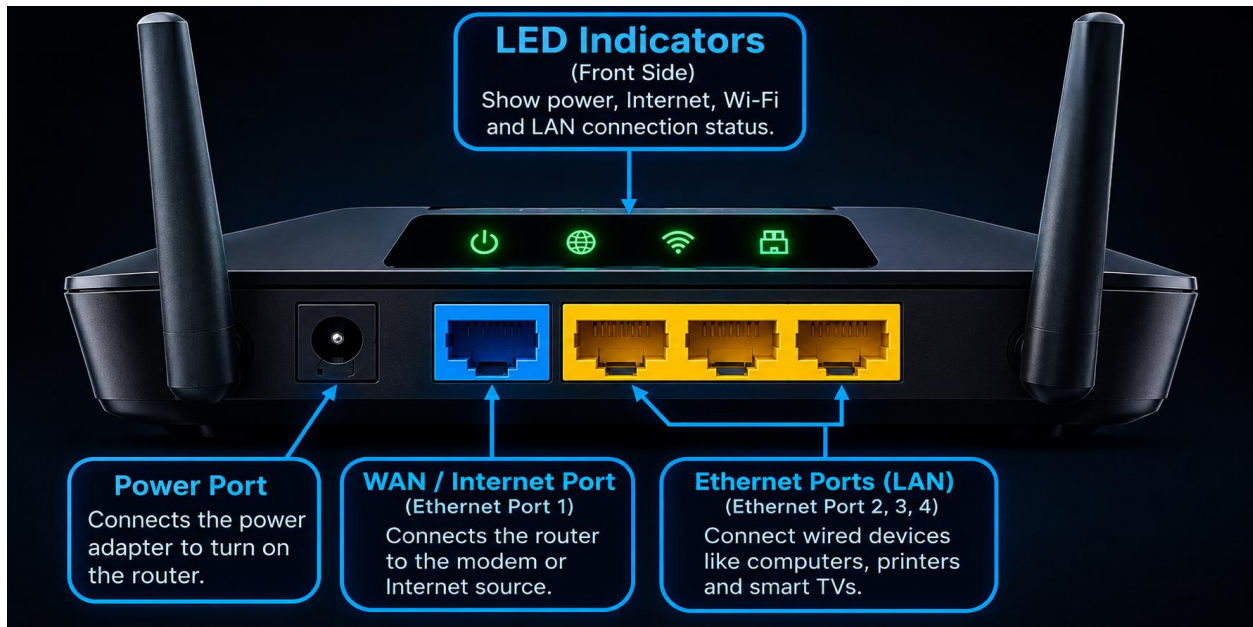
2 List 3 key differences between routers and switches in terms of their main functions and how they handle network traffic.

Hint: Think about how each device deals with IP addresses, MAC addresses, and the types of networks they connect.

Point	Router	Switch
Address Used	Uses IP Address	Uses MAC Address
Network Type	Connects different networks	Connects same network
Traffic Handling	Routes internet traffic	Forwards data between devices

3 Take a real or virtual image of a router and a switch (from your home, lab, or Google Images), and identify the following on each: Power port, at least 2 Ethernet ports, and any LED indicators. Annotate the image or describe their position and purpose.

◆ ROUTER



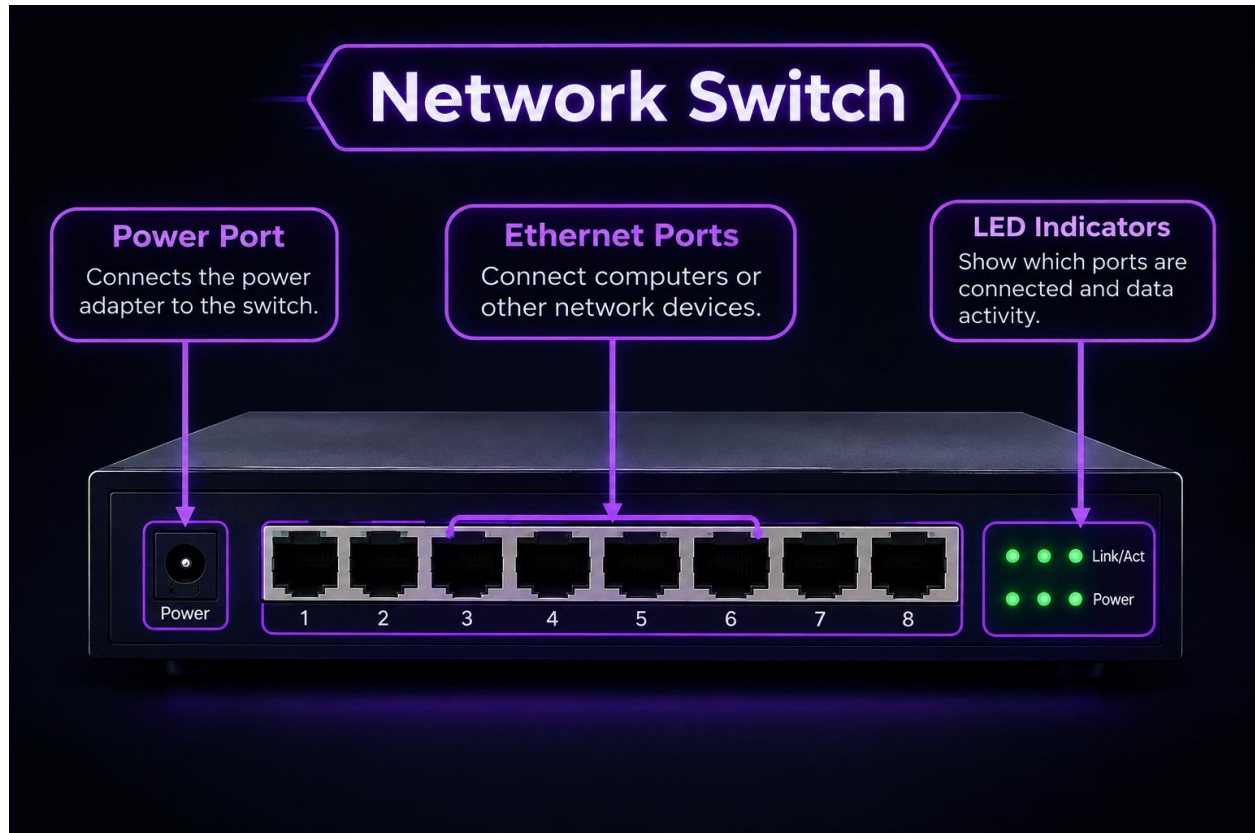
→ **Power Port**: Located on the **back side**. It is used to connect the **power adapter**.

→ **Ethernet Port 1 (WAN)**: Located on the **back side**. It connects the router to the **modem/Internet**.

→ **Ethernet Port 2 (LAN)**: Located on the **back side**. It connects a **computer or other wired device** to the network.

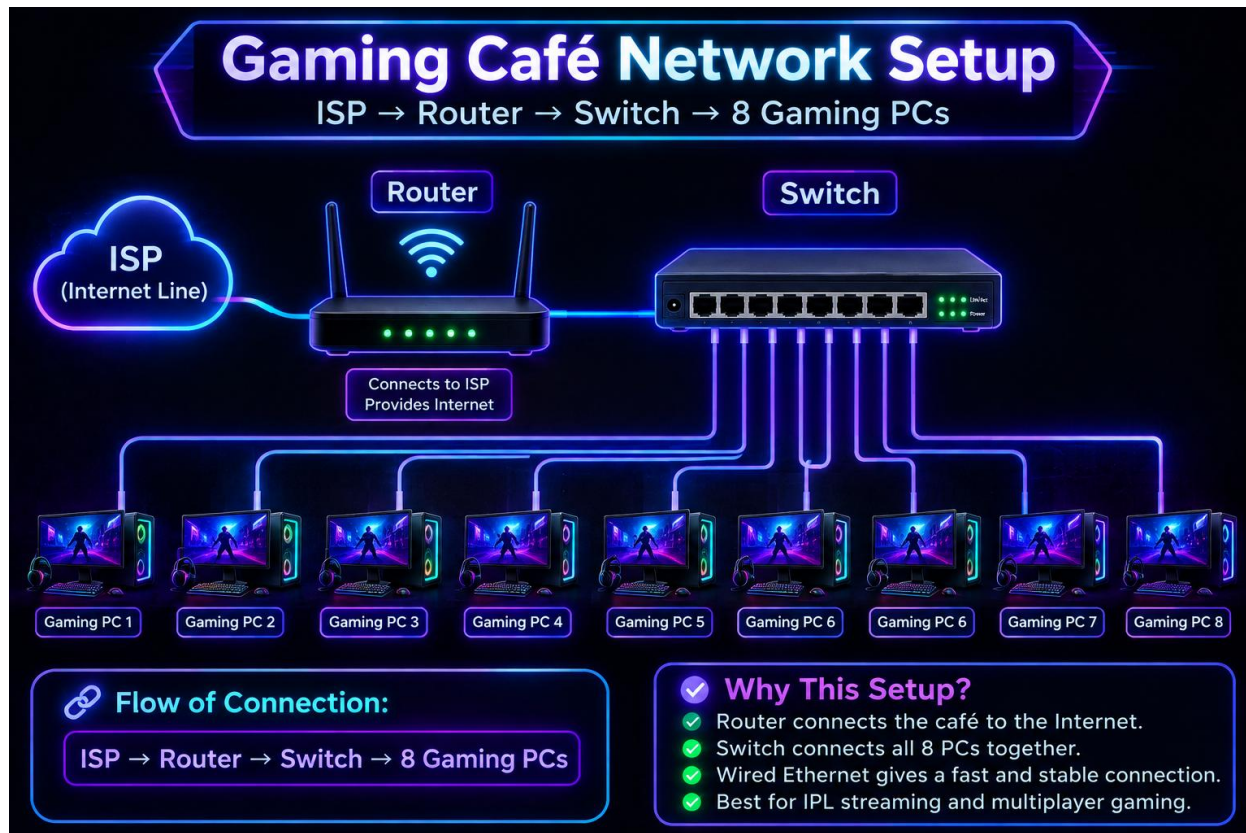
→ **LED Indicators**: Located on the **front side**. They show **Power, Internet, Wi-Fi, and connection status**.

◆ SWITCH



- **Power Port:** Located on the **back side**. It is used to connect the **power adapter**.
- **Ethernet Port 1:** Located on the **front side**. It connects a **computer or network device**.
- **Ethernet Port 2:** Located on the **front side**. It connects another **computer or network device**.
- **LED Indicators:** Located near the **Ethernet ports**. They show **connection and data activity**.

4 Imagine you are setting up a small gaming café network for IPL streaming and multiplayer matches. Write a short scenario explaining which device (router or switch) you would connect first to the ISP line, and how you would connect 8 gaming PCs to the network. Justify your choices.



Gaming Café Network

- First, I connect the **ISP line to the Router**.
- The **Router** provides Internet to the café.
- Then, I connect the **Router to the Switch** using an Ethernet cable.
- I connect **8 Gaming PCs to the Switch** using Ethernet cables.

- The **Switch connects all 8 PCs** to the network.
- Wired Ethernet gives a **fast and stable connection** for gaming.

➤ **Network Connection**

ISP → Router → Switch → 8 Gaming PCs

Justification

The **Router** connects the café to the Internet. The **Switch** connects all 8 gaming PCs to the network. This setup is **fast, stable, and good for IPL streaming and multiplayer gaming**.